SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Code of Conduct:

I K.ASWINI(192312397)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
K.ASWINI
Signature of the student

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*
Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
(b) Summarise the key objective of the article in your own words (150-200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used–size, features, source, and how it was prepared.
(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Article Title:

Machine Learning Based Mobile Applications for Autonomous Fertilizer Suggestion

Task 1: Article Summary

1(a) Full Citation, Domain and ML Problem

Full Citation

S. Raviraja, K. V. Raghavender, P. Sunagar, R. K. Ragavapriya, M. Jogendra Kumar, and V. G. Bharath, "Machine Learning Based Mobile Applications for Autonomous Fertilizer Suggestion," *Proceedings of the 2022 4th International Conference on Inventive Research in Computing Applications (ICIRCA)*, 2022, pp. 868–874.

DOI: 10.1109/ICIRCA54612.2022.9985721

The paper was published through IEEE in 2022, satisfying the requirement that the selected research article be published after 2019.

Domain/Application Area

The application domain is smart agriculture / precision agriculture.

Specific ML Problem

The paper addresses the problem of automatically recommending suitable fertilizer to farmers using environmental, soil, and crop-related information. The authors develop a machine-learning-based mobile application that predicts an appropriate fertilizer based on input parameters.

The study compares three supervised machine-learning techniques:

- K-Nearest Neighbour (KNN)
- Random Forest (RF)
- Decision Tree (DT)

The best-performing model is then selected for deployment in a mobile application.

1(b) Key Objective and Research Gap

The main objective of the research is to develop an automated fertilizer recommendation system that can help farmers select suitable fertilizers based on soil and environmental conditions. Traditional fertilizer selection can depend heavily on manual knowledge and may result in inappropriate fertilizer use. Excessive or unsuitable fertilizer application can reduce soil quality and cause environmental problems. Therefore, the authors investigate whether machine-learning models can provide a more convenient and automated method of fertilizer recommendation.

The research gap identified by the authors is the lack of easily accessible, automated fertilizer recommendation systems that can directly assist farmers using digital technology. The authors attempt to fill this gap by combining machine-learning classification with a smartphone application. Data containing environmental conditions, soil and crop type, and soil nutrients such as nitrogen, phosphorus, and potassium are processed and supplied to different ML models. KNN, Random Forest, and Decision Tree models are trained and compared. The best model is subsequently integrated into a mobile application developed using MIT App Inventor. The study therefore connects machine-learning prediction with a practical mobile-based agricultural application.

Task 2: Methodology Analysis

2(a) ML Techniques Used

The authors used three supervised machine-learning algorithms.

1. K-Nearest Neighbour (KNN)

KNN is a supervised learning algorithm that classifies a new data point according to the classes of its nearest training examples.

For fertilizer recommendation, the algorithm compares the input soil and environmental conditions with similar records in the training dataset and predicts the fertilizer associated with the nearest examples.

2. Decision Tree (DT)

A Decision Tree is a supervised learning algorithm that makes predictions through a sequence of decision rules.

For example, the model can make decisions based on:

Nitrogen → Phosphorus → Potassium → Moisture → Soil/Crop conditions → Fertilizer

Decision Trees are particularly relevant to the CO-4 syllabus because they perform inductive learning by learning decision rules from previously observed training examples.

3. Random Forest (RF)

Random Forest is an ensemble learning method that combines multiple decision trees. Each tree produces a prediction, and the forest combines the individual predictions to obtain the final classification.

In this paper, Random Forest performed better than the Decision Tree and KNN models. The authors therefore selected Random Forest for deployment in the mobile application.

Datasetz

The authors obtained the dataset from Kaggle. The dataset was provided in CSV format and was used for fertilizer prediction.

The attributes included:

Category	**Features**
Environmental conditions	Temperature, Humidity, Moisture
Soil/Crop information	Soil Type, Crop Type
Soil nutrients	Nitrogen (N), Potassium (K), Phosphorus (P)
Target	Fertilizer Name

The publicly reproduced version of the dataset contains 99 data records, with 8 input attributes and one fertilizer-label column. The attributes include temperature, humidity, moisture, soil type, crop type, nitrogen, potassium and phosphorus.

Data Preparation

The authors state that the data was collected in CSV format and processed to remove unwanted information. Categorical data was encoded into numerical form so that it could be processed by the machine-learning models. The complete dataset was divided into:

- 80% training data
- 20% testing data

The trained models were then evaluated using classification metrics.

2(b) Mapping to CO-4 Topics

Topic	Connection with Article
Inductive Learning	The models learn fertilizer-prediction patterns from previously observed training examples.
Decision Trees	Decision Tree is one of the three algorithms directly evaluated.
Statistical Learning	KNN and Random Forest are supervised statistical/machine-learning approaches used for classification.
Reinforcement Learning	Not directly used because the system learns from labeled examples rather than rewards and environmental interaction.

Methodological Strength

One major strength is that the authors compare multiple machine-learning algorithms rather than relying on only one model. KNN, Decision Tree, and Random Forest are evaluated using common classification measures. This provides a basis for selecting the most suitable model for deployment.

Methodological Limitation

A major limitation is the small dataset and relatively simple evaluation procedure. The study uses an 80:20 train-test division, but there is no strong evidence of repeated cross-validation or testing on an independent real-world dataset. Therefore, the reported performance may not fully represent performance under different agricultural conditions.

Task 3: Results and Critical Evaluation

3(a) Key Results

The authors compared KNN, Random Forest, and Decision Tree.

Accuracy Comparison:

Model	Accuracy
KNN	80%
Random Forest	90%
Decision Tree	85%

The Random Forest model achieved the highest accuracy of 90%, followed by Decision Tree at 85%, while KNN achieved 80%.

Classification Performance

The paper reports that Random Forest also achieved the strongest overall classification performance. Its precision reached 96%, while KNN had lower performance, including an F1-score of approximately 81%. The authors concluded that Random Forest outperformed the other two models.

Model	Accuracy	General Performance
KNN	80%	Lowest
Decision Tree	85%	Moderate
Random Forest	90%	Best

The Random Forest model was therefore selected and deployed in the mobile application.

3(b) Critical Evaluation

The reported results show that Random Forest performs better than KNN and Decision Tree. The improvement from 80% to 90% accuracy is meaningful for a preliminary fertilizer recommendation system.

However, the results should be interpreted carefully.

Dataset Bias

The dataset was obtained from Kaggle rather than being collected extensively from different geographical agricultural regions. Soil properties and fertilizer requirements can vary considerably according to location, crop variety, climate, and farming practices.

Therefore, a model trained on a limited dataset may not generalize equally well to all farms.

Evaluation Bias

The authors used an 80:20 train-test split. A single split can sometimes produce a performance estimate that depends on the particular division of the data.

A stronger evaluation would use:

- 5-fold or 10-fold cross-validation
- Repeated train-test experiments

- Independent external datasets
- Testing using real farm data

Realism of Metrics

An accuracy of 90% is promising, but accuracy alone does not guarantee that the model will make safe recommendations for every fertilizer class. Precision, recall, F1-score and confusion matrices should also be considered, particularly if some fertilizer classes have fewer examples.

Additional Experiments

The study could be strengthened by:

1. Using a much larger dataset.
2. Collecting data from different geographical regions.
3. Performing k-fold cross-validation.
4. Testing additional algorithms such as SVM, Naïve Bayes, XGBoost and Gradient Boosting.
5. Using real-time soil-sensor measurements.
6. Evaluating the model during different agricultural seasons.
7. Measuring fertilizer recommendation errors in real farms.

3(c) Real-World Applicability

The proposed system can be deployed in smart farming and precision agriculture.

Potential applications include:

- Mobile fertilizer recommendation systems
- Smart agriculture platforms
- Agricultural advisory applications
- IoT-based farming systems

- Soil monitoring systems
- Precision fertilizer management
- Farmer decision-support systems

The paper demonstrates practical deployment by integrating the selected Random Forest model into a mobile application developed using MIT App Inventor.

Challenges in Industrial Scaling

1. Data Availability

Large and diverse agricultural datasets are required. Data should represent different soils, crops, climates, regions and seasons.

2. Sensor Accuracy

If the application uses sensors to obtain soil information, inaccurate sensor measurements can produce incorrect recommendations.

3. Model Generalization

A model trained using one region's data may not perform equally well in another region.

4. Connectivity

Farmers in rural areas may have limited or unreliable internet access. Offline prediction capability would therefore be useful.

5. Ethical and Environmental Issues

Incorrect fertilizer recommendations could cause economic loss or environmental damage. Therefore, the system should be treated as a decision-support tool and should ideally provide confidence levels and expert verification.

6. Computational Requirements

Although Random Forest can be used on relatively lightweight systems, large-scale systems with continuous sensor data may require cloud or edge-computing infrastructure.

Task 4: Reflection and Learning Outcome

4(a) Three Key Insights

Insight 1: Machine Learning Can Perform Inductive Learning

The article demonstrates how a model can learn general fertilizer-prediction patterns from previously observed examples.

This connects directly to Inductive Learning, where a general rule is learned from specific training examples.

Learning: I understood that machine learning does not require manually writing every decision rule. Instead, the model can learn relationships from existing data.

Insight 2: Decision Trees Are Useful for Classification

The paper directly applies a Decision Tree to fertilizer classification.

This connects to the Decision Tree topic in CO-4. A Decision Tree divides the input data using feature-based decisions until a final class is obtained.

Learning: I understood how soil and environmental parameters can be converted into a sequence of decisions that ultimately produces a fertilizer recommendation.

Insight 3: Ensemble Learning Can Improve Prediction

Random Forest performed better than the individual Decision Tree and KNN models, achieving 90% accuracy.

This connects to Statistical Learning and Ensemble Methods.

Learning: I learned that combining multiple decision trees can produce a more robust prediction than using a single tree. This explains why Random Forest can outperform a basic Decision Tree.

4(b) Proposed Extension / Novel Experiment

Proposed Experiment: IoT-Based Real-Time Fertilizer Recommendation

If I were extending this research, I would develop an IoT-based real-time fertilizer recommendation system.

Instead of manually entering soil parameters into the mobile application, sensors could continuously collect:

- Soil moisture
- Temperature
- Humidity
- Soil pH
- Nitrogen
- Phosphorus
- Potassium

The sensor data could be transmitted to a mobile application or cloud platform. Multiple ML algorithms such as Random Forest, Decision Tree, SVM and XGBoost could then be compared using cross-validation.

The system could also provide a confidence score with every recommendation.

Justification

This enhancement is feasible because soil sensors and mobile/IoT platforms are already widely available. Real-time measurements would reduce dependence on manually entered data and allow the recommendation system to respond to changing soil conditions.

Expected Impact

The proposed system could:

- Improve fertilizer recommendation accuracy.
- Reduce excessive fertilizer usage.
- Reduce environmental pollution.
- Improve crop productivity.
- Reduce fertilizer costs.
- Support precision agriculture.
- Provide farmers with faster decision-making assistance.

Overall Critical Review

The article presents a useful application of supervised machine learning in agriculture. Its main contribution is the integration of machine-learning-based fertilizer prediction with a mobile application. The comparison of KNN, Decision Tree, and Random Forest demonstrates the importance of evaluating different algorithms before deployment.

The reported results show that Random Forest achieved the best accuracy of 90%, compared with 85% for Decision Tree and 80% for KNN.

However, the research would be stronger with a larger and more diverse dataset, repeated cross-validation, independent field testing, and real-time sensor data. The study provides a good practical example of inductive learning, Decision Trees, and statistical machine learning and demonstrates how CO-4 concepts can be applied to a real-world agricultural problem.

Conclusion

The reviewed research successfully demonstrates how machine-learning techniques can be applied to fertilizer recommendation. The authors compare KNN, Decision Tree, and Random Forest models and identify Random Forest as the best-performing model with 90% accuracy. The selected model is subsequently integrated into a mobile application, making the research practically relevant.

From the perspective of CO-4, the paper provides a clear example of inductive learning and supervised classification, particularly Decision Trees and Random Forest. Although the system shows promising results, larger datasets, stronger validation methods, real-world field trials, and IoT-based data collection would improve its reliability and scalability.

Article Link

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View the article on IEEE / DOI

Reference

Raviraja, S., Raghavender, K. V., Sunagar, P., Ragavapriya, R. K., Kumar, M. J., & Bharath, V. G. (2022). *Machine Learning Based Mobile Applications for Autonomous Fertilizer Suggestion*. Proceedings of the 4th International Conference on Inventive Research in Computing Applications (ICIRCA), 868–874. DOI: 10.1109/ICIRCA54612.2022.9985721.